DATA SCIENCE & ANALYTICS · SESSION 08

RFM Analysis & Customer Segmentation

Use Case 2: turning raw transactions into customer segments you can act on

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2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

Session Agenda

01 **The Customer Question**
From products to people

02 **Recency, Frequency, Monetary**
The three numbers that describe a customer

03 **Scoring with qcut**
Turning raw values into 1–4 tiers

04 **Assigning Segments**
From scores to Champions and At-Risk

05 **Profiling Segments**
Reading each group's business meaning

06 **Workshop Block**
Segment the customers and name them

01

PART 01

The Customer Question

The same retailer now wants to understand its customers: who is most valuable, who is slipping away, and how to treat each group differently.

Use Case 2 shifts the unit of analysis from the product to the person

From products to people

New question	Not 'which product sells?' but 'who are our best customers, and who is about to leave?'
Why it matters	Keeping a good customer is far cheaper than winning a new one. Segmentation tells you who to keep and how.
The data	The same 5,000 transactions, now grouped by customer_id — about 500 customers with a purchase history each.
The output	A handful of named, actionable segments — Champions, Loyal, At-Risk — each with a different treatment.

Source: DS Course, Session 8, 2026

02

PART 02

Recency, Frequency, Monetary

RFM compresses a whole purchase history into three numbers that, together, describe how valuable and how engaged each customer is.

RFM: three numbers that summarize any customer

The three dimensions

Recency

Days since their last purchase. Lower is better — a recent buyer is far more likely to buy again.

Frequency

How many times they've purchased. Higher means a stronger, stickier habit.

Monetary

Total spend. Higher means more revenue at stake. Compute all three with one groupby on customer_id.

PYTHON · RFM

```
snapshot = df['date'].max()

rfm = df.groupby('customer_id').agg(
    recency=('date',
        lambda s: (snapshot - s.max()).days),
    frequency=('order_id', 'count'),
    monetary=('revenue', 'sum'),
)
```

One groupby turns a history into three numbers.

Scoring with qcut: turn raw values into comparable 1–4 tiers

Make the three comparable

Why score

Recency in days and monetary in baht aren't comparable. Scoring puts all three on the same 1–4 scale.

qcut

Splits each dimension into quartiles — equal-sized groups. Each customer gets a 1–4 on R, F, and M.

Mind direction

For recency, low days = best, so reverse the labels. Frequency and monetary score high-is-best.

PYTHON · qcut SCORING

```
rfm['R'] = pd.qcut(rfm['recency'], 4,
    labels=[4, 3, 2, 1])   # low days = best
freq_rank = rfm['frequency'].rank(method='first')
rfm['F'] = pd.qcut(freq_rank, 4,
    labels=[1, 2, 3, 4])
rfm['M'] = pd.qcut(rfm['monetary'], 4,
    labels=[1, 2, 3, 4])
```

qcut makes equal-sized, comparable tiers.

03

PART 03

Assigning Segments

Combine the three scores into a single label that maps each customer to a business archetype with an obvious next action.

Score combinations become named segments — each with its own playbook

From numbers to names

Champions	High on all three — recent, frequent, big spenders. Reward and retain them; they're the core of the business.
Loyal	Frequent and high-value but less recent. Nurture before they drift; a small nudge keeps them engaged.
At-Risk	Once valuable, now long absent. High monetary, poor recency. Win them back before they're gone for good.
Lost / New	Lost: low on everything — low-cost reactivation only. New: recent but unproven — onboard and encourage a second buy.

Source: DS Course, Session 8, 2026

04

PART 04

Profiling Segments

A segment is only useful once you can describe it — its size, its value, and the behaviour that defines it — in business terms.

Profile each segment: size, value, and behaviour

Make segments legible

Size

How many customers fall in each segment. A tiny Champions group and a huge Lost group is a real warning.

Value

Average and total monetary per segment. Confirms Champions really are where the revenue concentrates.

Visualize

A bar of segment sizes and revenue side by side tells the whole story to a non-technical reader at a glance.

PYTHON · PROFILE

```
profile = rfm.groupby('segment').agg(
    customers=('segment', 'count'),
    avg_monetary=('monetary', 'mean'),
    avg_recency=('recency', 'mean'),
).round(0)

profile.sort_values('avg_monetary',
                    ascending=False)
```

A segment you can't profile, you can't act on.

05

PART 05

Profiling Segments

Tie each profile back to a concrete action so the segmentation drives marketing, not just a pretty chart.

Segmentation earns its keep only when each segment gets a different action

From profile to playbook

Champions

VIP treatment, early access, referral asks. Protect this group above all — losing one is expensive.

At-Risk

A targeted win-back: a personal offer, a 'we miss you' touch. The highest-ROI campaign you can run.

Different budgets

Spend to retain Champions and recover At-Risk; spend little on Lost. Equal treatment wastes money.

Measure

After acting, re-run RFM next quarter. Did At-Risk shrink? Did Champions grow? The loop closes here.

Source: DS Course, Session 8, 2026

06

PART 06

Workshop Block

Compute RFM for the customer base, assign segments, and name them — then state the one action you'd take for each.

By the end of this block, every student has a named, profiled customer segmentation

4 Steps · 35 Minutes

Build RFM	groupby customer_id for recency, frequency, monetary. Confirm ~500 customers and sensible ranges.
Score	qcut each dimension into 1–4 tiers, respecting direction for recency.
Segment & name	Combine scores into 4–5 named segments. Champions, Loyal, At-Risk, Lost, New.
One action each	Profile the segments and write the single action you'd take for each. Profile plus actions is your deliverable.

Source: Workshop, Session 8, 2026

KEY TAKEAWAY

Three numbers — recency, frequency, money — turn customers into a strategy.